SARTHI

AI-Assisted Investment Decision Support System

Project Summary & Strategy Document

Version 2.0

Overview

Sarthi *(meaning guide or charioteer in Hindi)* is a personalised web application that helps long-term retail investors in India make more informed, rational investment decisions. It scans trusted financial sources every morning, identifies events that matter, reasons through their implications, and delivers a short, well-reasoned briefing tailored to each user's financial profile.

> *Core philosophy: Sarthi guides, it never decides. Every final decision remains with the user.*

Objective

Sarthi aims to close the **rationality gap** that causes retail investors to make emotionally driven decisions during market uncertainty. The system processes financial events from credible institutional sources, applies a structured multi-step reasoning pipeline, and delivers personalised, well-reasoned investment briefings — filtered for relevance to each user's specific profile and risk tolerance.

> *The goal is not to replace human judgment, but to sharpen it.*

The Problem It Solves

Retail investors are overwhelmed by financial news. They cannot easily distinguish confirmed, actionable events from noise. Existing tools either give raw news with no reasoning, or black-box price predictions with no explanation — neither is clear, personalised, or conservative enough for a long-term investor.

Who It's For

Long-term retail investors in India — from first-time investors to experienced ones — who want structured, well-reasoned guidance rather than tips or predictions. The app is designed to be accessible regardless of financial literacy level.

Core Philosophy

Conservative by default

Biased toward No Action. A recommendation requires sufficient source confirmation and a confidence threshold. Staying quiet when evidence is weak is a feature, not a failure.

Reasoning-first always

Every output tells the user not just what to consider but why — with historical context, source clarity, and a fully auditable reasoning chain. No black box outputs.

Human authority

Sarthi never executes anything. It never gives tips. It provides structured analysis and the user decides.

Candid about limitations

The system states its uncertainty explicitly. If confidence is low, it says so. If drawing on general reasoning rather than verified data, it says so.

How It Works: The Pipeline

When the user opens the app or triggers the morning brief, the system runs a full **7-step pipeline**. Every event must pass through all steps before reaching the user.

Step	Name	What Happens
1	**Scan**	Trusted sources scanned top-down by credibility tier. Higher tiers always checked first.
2	**Classify**	Each event placed into one of four categories — policy/regulatory, earnings/corporate, macro/economic, or geopolitical.
3	**Deduplicate**	Same event across multiple sources clustered into one object. Source count tracked as confirmation signal.
4	**Confirm**	Each event checked against source-tier confirmation rules. Unconfirmed events go no further.
5	**Reason**	AI reasons through sector and portfolio implications. Flags whether curated data or general reasoning is used.
6	**Constrain**	Constraint layer checks confidence score, profile compatibility, and drawdown tolerance. Weak outputs blocked.
7	**Output**	Verdict delivered with full reasoning chain, confidence score, and explicit caveat.

> ***Two exit points exist within the pipeline.*** *Events failing the Confirm step are dropped entirely. Events failing the Constrain step are downgraded or blocked. Staying quiet when evidence is weak is a feature, not a failure.*

A Practical Example

On the morning of the Union Budget announcement, the Finance Minister raises cigarette excise duty by 15%.

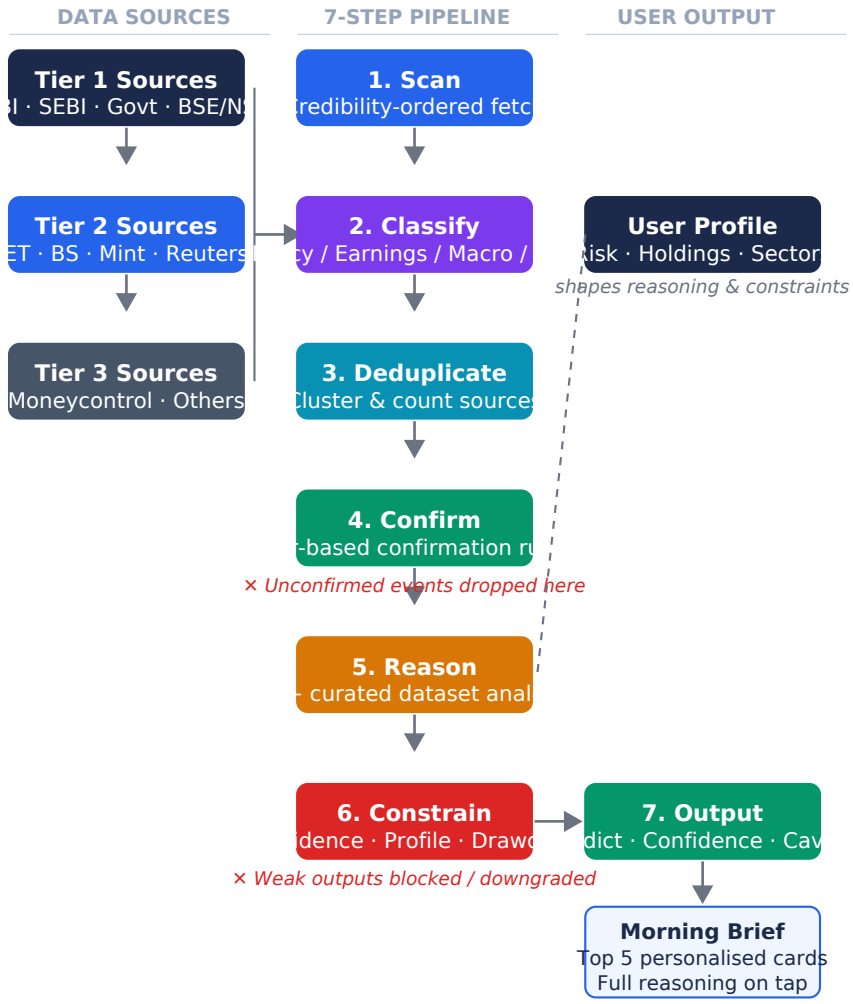
1. **Scan** — Official Budget document picked up from the Ministry of Finance website (Tier 1).
2. **Classify** — Categorised as Policy and Regulatory, triggering the regulatory reasoning chain.

3. **Deduplicate** — Same announcement detected across Economic Times and Mint; clustered into one event object with source count of 3.
4. **Confirm** — Single Tier 1 source is sufficient. Event passes immediately.
5. **Reason** — Higher excise duty → increased production cost → near-term margin pressure for ITC. Qualitative analogy drawn from post-GST period (curated dataset).
6. **Constrain** — For a user holding ITC with moderate risk profile, confidence is 68% — above threshold. For a user with no FMCG exposure, event is downgraded to the Explore feed.
7. **Output** — Morning brief card: *ITC — Sell/Reduce, 68% confidence. Cigarette excise duty raised → FMCG pricing pressure expected short-term. [Curated dataset].*

A user with no tobacco holdings never sees this card. A user who has marked tobacco as an ethical avoidance sees it only in the Global Feed, with no verdict attached.

System Architecture

The diagram below shows the complete end-to-end flow — from data ingestion through to the personalised output delivered to the user.



Source Hierarchy

One of the core research contributions of this system. Most tools treat all sources equally. Sarthi does not.

Tier	Sources	Confirmation Rule
Tier 1 Primary	RBI, SEBI, Government of India, BSE/NSE official announcements	Single source sufficient. These institutions are the event.
Tier 2 Enrichment	Economic Times, Business Standard, Mint, Reuters India, Bloomberg India	Two sources required for Tier 2-originated stories.
Tier 3 Corroborating	Moneycontrol and smaller outlets	Never sole source. Useful for early detection only.

Honest Historical Analogy

A deliberate design decision based on a known LLM weakness. LLMs cannot reliably produce numeric historical claims — saying "*this happened 3 of 5 times*" sounds authoritative but risks hallucination.

Sarthi uses **qualitative analogies** instead, grounded in a curated dataset of **20–30 well-documented historical events** — including Demonetisation, GST implementation, RBI rate cycles, and the COVID market crash and recovery.

Every historical reference states clearly whether it comes from the curated dataset or general AI reasoning, with different confidence weights applied to each. This is verifiable, grounded, and academically defensible.

Personalisation Model

Users choose how much to share. The system is explicit about the accuracy tradeoff at each level.

Level 1 — Preferences only

Risk tolerance, investment horizon, sectors to include or avoid. General sector-level guidance. Fully private.

Level 2 — Rough holdings

Broad sector and cap-size exposure, no specific stocks needed. More targeted output.

Level 3 — Specific holdings

Enter main stocks manually or upload a CSV. Most personalised output. All data processed locally and never sent to any server.

Tech Stack

Layer	Technology	Reason
Frontend	Next.js	Web-first, SSR, clean routing that maps to Sarthi's screens
Backend	Python / FastAPI	AI-native, pipeline-friendly, auto-generates API documentation
Database	Supabase (PostgreSQL)	Structured data, built-in auth, generous free tier for MVP
AI / LLM	Claude API (Anthropic)	Strong instruction-following; respects strict reasoning constraints
AI Pipeline	Custom (no framework)	Full traceability — no abstraction hiding the reasoning chain

Frontend — Next.js

Built with Next.js for fast server-side rendering and clean file-based routing that maps directly to Sarthi's screens.

Backend — Python / FastAPI

The core 7-step pipeline runs on a Python backend via FastAPI. Python is the natural choice — all major LLM libraries are Python-first.

Database — Supabase (PostgreSQL)

User profiles, event records, and the curated historical dataset are stored in Supabase. Level 3 portfolio data is processed entirely in the browser and never written to the database.

AI Layer — Claude API (Anthropic)

Claude is selected specifically because Sarthi's Honest Historical Analogy model requires an LLM that follows strict constraints against fabricating numeric historical claims.

Pipeline Architecture — Custom

The 7-step pipeline is a custom Python module. Abstraction frameworks obscure the reasoning chain, which conflicts with Sarthi's commitment to full traceability.

User Interface & Experience

Morning Brief (Main Screen)

Top 5 personalised event cards, already reasoned and ranked by relevance to the user's profile.

> **ITC — Sell/Reduce, 68% confidence**
> Cigarette excise duty raised in Union Budget → FMCG pricing pressure expected short-term. Similar to post-GST period where ITC saw margin compression before recovering. [Curated dataset].
> *Tap to open full reasoning chain.*

Additional Features

Global Feed

Unpersonalised view of what is happening across Indian and global markets today. No verdicts — the user contextualises it themselves.

Explore

Events that did not cross the personalisation threshold. Shown with a clear label: "Low relevance to your current profile. General analysis only."

Profile & Settings

Edit any part of the profile at any time.

The Full Reasoning Chain

- **Event detected** — What happened, which source tier confirmed it, classification category, and how many sources.
- **Why it matters** — Macro or sector implication, reasoning explained plainly.
- **Historical reference** — Qualitative analogy, flagged as curated dataset or general reasoning.
- **Profile fit** — Why this does or does not match the user's specific constraints.
- **Verdict** — Buy/Add, Sell/Reduce, Monitor/Hold, or No Action.
- **Caveat** — Key uncertainty or limitation stated explicitly.

Chatbot Layer (Planned)

Two distinct chatbots planned for later phases, kept separate by design.

General financial assistant

Always available, personalised to the user's profile and portfolio. Answers questions about financial concepts, portfolio overlap, sector concentration, and historical context. Education and analysis only — never tips.

Event-specific assistant

Appears contextually when viewing an event card. Loaded with that event's full context and the user's profile. The user can probe the reasoning, ask follow-up questions, and challenge the logic. Scoped strictly to that event.

> *The separation is intentional. They serve different cognitive modes — keeping them separate makes both more reliable and more useful.*

Onboarding & Profile Setup

Seven screens, progressive and conversational. Designed to earn trust gradually before asking sensitive questions.

1. **Name and age band** — Low friction entry.
2. **Income situation and engagement** — Shapes tone and explanation depth.
3. **Investment profile** — Experience level, goal, horizon, asset comfort.
4. **Risk profiling** — Three-step behavioural process using real rupee amounts to reveal actual behaviour vs. stated tolerance.
5. **Portfolio disclosure** — User chooses Level 1, 2, or 3. Accuracy tradeoff explained.
6. **Sector preferences** — Including avoidances for ethical, religious, or personal reasons.
7. **Full profile summary** — Review in plain English before confirming.

Academic Contributions

- **Source credibility architecture** — Financial sources ranked into three tiers by institutional authority. Confirmation rules tied to source tier, not just source count.
- **Event processing pipeline** — Every event passes through explicit classification, deduplication, and confirmation steps. Nothing reaches the user without passing all three gates.
- **Constrained AI reasoning** — AI handles language understanding; a separate structured layer manages thresholds, confidence scoring, and profile compatibility checks.
- **Honest historical analogy** — Uses a curated dataset of well-documented Indian market events for qualitative analogies instead of unverifiable numeric claims. Every reference is flagged as curated data or general AI reasoning.
- **Revealed vs. stated preference risk profiling** — Cross-checks stated risk tolerance against revealed behaviour using real rupee-value scenarios, drawing on Kahneman and Thaler.
- **Personalisation under uncertainty** — The system is upfront about accuracy tradeoffs. Users who share less get less precise output — and the app tells them so directly.
- **Full traceability as a first-class feature** — Every verdict comes with a step-by-step reasoning chain showing what triggered it, which sources confirmed it, what analogy was used, and why it was flagged.

> ***AI output disclaimer:*** *All reasoning and verdicts produced by the system are AI-generated and may occasionally be incorrect or incomplete. Users are expected to apply their own judgement before acting on any output.*

Strategic Deep Dive: Addressing the Rationality Gap

99% of daily financial news is noise. Sarthi is not a news aggregator — it is a **noise-cancellation engine** designed to solve the rationality gap that causes retail investors to panic-sell during structural market shifts.

1. The Structural Shift Thesis

Policy and regulatory changes are structural, not transient. Sarthi ignores market chatter and focuses strictly on events that alter the **long-term fundamental math of a sector**. Tier 1 sources are self-confirming — they do not need media interpretation to be valid.

2. Solving the AI Hallucination Problem

Standard LLMs hallucinate numeric financial claims to sound authoritative. Sarthi explicitly bans numeric historical claims, using a curated dataset of 20–30 major events for qualitatively grounded analogies. Every output flags whether reasoning is based on verified data or general AI reasoning.

3. Behavioural Finance vs. Passive Tracking

Standard portfolio analysers assume a perfectly rational investor — which behavioural finance proves is false. Sarthi incorporates the work of **Kahneman and Thaler** to manage emotional

response, cross-checking stated risk tolerance against revealed behaviour. Unlike most fintech apps, Sarthi is **conservative by default**.

4. Technical Architecture: The Constraint Layer

Sarthi uses a **hybrid architecture** — AI for Natural Language Understanding, and a structured constraint layer for thresholds, profile compatibility, and drawdown tolerance. If a reasoning chain does not pass these checks, the output is blocked or downgraded. The user always remains the decision-maker.

5. Privacy-First Personalisation

Level 3 holdings are processed locally in the browser and never sent to a server, allowing the most personalised output possible while maintaining absolute data sovereignty.

Fundamental Differentiators

Compared to	They do	Sarthi does
News aggregators	Prioritise volume and speed, overwhelming users	Filter for policy changes that alter sector fundamentals via a 7-step pipeline
Portfolio analysers	Provide data points without contextual reasoning	Provide a full reasoning chain incorporating behavioural finance principles
Trading apps	Encourage high-frequency engagement and constant action	Default to no action when evidence is weak — silence is a successful output
Black-box AI tools	Offer opaque predictions with no explainable logic	Govern AI output with a constraint layer; every output is flagged and traceable

Unique Selling Propositions

Rationality gap solution

Engineered as a noise-cancellation engine to bridge the gap between market data and human psychology. Focuses on mitigating the panic-selling reflex, not contributing to information overload.

Tiered source rigour

A three-tier hierarchy that prioritises institutional accuracy over media speed. Tier 1 sources are treated as self-confirming events.

Hallucination-proof historical analogies

Strictly banned from making numeric historical claims. Uses a curated dataset of 20–30 major events for qualitatively grounded, verifiable analogies.

Behavioural cross-checking

Does not take stated risk tolerance at face value. Cross-checks against revealed behaviour using real rupee-value loss scenarios.

Privacy-first edge personalisation

Level 3 holdings processed entirely within the user's browser. Sensitive financial data is never sent to a central server.

Conservative verification gates

Every AI-generated reasoning chain must pass a structured constraint layer. Outputs that don't meet thresholds are intentionally blocked or downgraded.

Future Scope

Chatbot layer

Two conversational assistants — a general financial assistant and an event-scoped assistant. Neither will give tips; both are scoped to education, analysis, and reasoning support.

Native mobile application

Dedicated iOS and Android app enabling push notifications for time-sensitive structural events.

Backtesting engine

Retrospective layer testing historical analogies against documented outcomes, progressively refining the curated dataset.

Expanded historical dataset

Broader sector-specific events, global macro shocks with India-specific impact, and RBI policy cycle granularity.

US and global market support

Extending coverage to US equities, global ETFs, and cross-border macro events.

Automated background scheduling

Background scheduler running the morning scan automatically before market hours.

Paywalled source integration

Licensed access to Bloomberg and similar Tier 2 sources to raise the quality ceiling of the enrichment layer.

Cross-device portfolio persistence

Encrypted sync model for Level 3 holdings stored server-side in a user-controlled format.

Scope & Limitations

Out of Scope (MVP)

Trade execution, price prediction, automated background scheduling, native mobile app, backtesting engine, US market coverage, and paywalled source access.

Known Limitations

- Historical analogy quality depends on the curated dataset size (20–30 events).
- Event classification by AI is imperfect — the visible pipeline allows user judgment.
- No empirical backtesting yet.
- Portfolio data at Level 3 stays in-browser with no cross-device persistence.

- Real-time scanning depends on web search availability.
- **AI output disclaimer** — All reasoning and verdicts are AI-generated and may occasionally be incorrect or incomplete. Users are expected to apply their own judgement before acting on any output.

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